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26 AUGUST 1953
SUPERSEDING
JAN-N-589
30 APRIL 1948

MILITARY SPECIFICATION

THICKENER, INCENDIARY OIL, M1

This specification has been approved by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of thickener for incendiary fuels.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-T-191 —Terne-Plate (Long Ternes).
RR-S-366 —Sieves; Standard, Testing.
TT-C-595 —Colors; (for) Ready-Mixed Paints.
UU-P-641 —Paper, Writing.
VV-L-791 —Lubricants, Liquid Fuels, and Related Products; Methods of Inspection, Sampling, and Testing.

MILITARY

JAN-L-73 —Lacquer, Enamel, Lusterless.
JAN-E-74 —Enamel, Lusterless, Quick-Drying.
JAN-P-106 —Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
JAN-C-490 —Cleaning and Preparation of Ferrous Metal Surfaces for Organic Protective Coatings.
JAN-I-711 —Incendiary Oils, Consistency of; Test Procedures.
MIL-E-10687 —Enamel; Lusterless, Quick-Drying (For Ammunition).

STANDARDS

MILITARY

MIL-STD-129—Marking of Shipments.
MS-602 —Test Solvent (for Fuel Thickeners).

DRAWINGS

CHEMICAL CORPS

D99-17-1 —Container — Assembly, Detail, and Bill of Material.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 **Other publications.**—The following publication, of the issue in effect on date of invitation for bids, unless otherwise stated, forms a part of this specification:

INTERSTATE COMMERCE COMMISSION

Regulations for Transportation of Ex-

270721—53—(222)

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plosives and Other Dangerous Articles, etc.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Material.—The contractor shall certify that the material is a coprecipitated mixture of basic aluminum soaps of naphthenic acid, oleic acid, and coconut fatty acids and that the material was manufactured subsequent to the date of the contract or order (see 4.5 and 6.3).

3.2 Inhibitor.—The thickener shall contain sufficient alpha-naphthol as a stabilizer to develop a color equal to or darker than a color standard containing 0.2-percent alpha-naphthol when tested as specified in 4.4.1.

3.3 Moisture.—The thickener shall contain not more than 0.8 percent nor less than 0.4 percent moisture when tested as specified in 4.4.2.

3.4 Granulation.—Not more than 0 percent of the material shall be retained on a No. 6 standard sieve and not less than 75 percent shall be retained on a No. 40 standard sieve when tested as specified in 4.4.3.

3.5 Sulfate.—The thickener shall contain not more than 3.0 percent sulfate when tested as specified in 4.4.4.

3.6 Gel formation.—All 8-percent gels, prepared and tested as specified in 4.4.5.2, shall perceptibly thicken in not more than 10 minutes as evidenced by the disappearance of the vortex around the stirrer shaft. The gels so formed shall be of uniform consistency and free from lumps.

3.7 Gel consistency.

3.7.1 *Four-percent gel.*—All 4-percent gels, prepared and tested as specified in 4.4.6.2 shall be uniform and free from lumps and shall have a consistency of not less than 100 gm. nor more than 200 gm. The average of the two consistencies for each of the two specimens tested shall differ not more than 50 gm.

3.7.2 *Eight-percent gel.*—All 8-percent gels prepared and tested as specified in 4.4.6.3 shall be uniform and free from lumps and shall have a consistency of not less than 550 gm. nor more than 750 gm. The average of the two consistencies for each of the two specimens tested shall differ not more than 80 gm.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.—When the thickener is manufactured by a batch process (see 6.5), each batch shall constitute a lot. If manufactured by a continuous process, a lot shall consist of the production from not more than 24 hours continuous operation.

4.2 Sampling.—The Government inspector shall take a 1-pound representative specimen from each of five containers selected at random from each lot. Each specimen shall be sealed in a separate, clean, dry, glass bottle, which shall be labelled to identify the lot represented (see 6.9).

4.3 Inspection.—The inspector shall inspect the packaging, packing, and marking for compliance with section 5.

4.4 Tests.—The number of specimens tested from each lot shall be as specified in table I. Reagent grade chemicals shall be used throughout unless otherwise specified.

4.4.1 *Inhibitor content.*—Test one specimen from each shift's production for inhibitor content. If the specimen fails to comply with 3.2, all material precipitated in the shift represented shall be rejected. Reagents and pro-

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cedures for the test for inhibitor content are as follows:

TABLE I.—*Number of specimens to be tested per lot.*

Test	Paragraph	No. of specimens to be tested
Inhibitor	4.4.1	(see 4.4.1)
Moisture	4.4.2	1
Granulation	4.4.3	1
Sulfate	4.4.4	1
Gel formation	4.4.5	5
Gel consistency:		
4-percent gel	4.4.6	2
8-percent gel	4.4.6	2

4.4.1.1 *Mixed solvent.* — Mix by volume: five parts iso-octane, five parts toluene, and three parts absolute ethanol.

4.4.1.2 *Reagent.*—Dissolve 4.0 gm. sodium bromide and 0.10 gm. cupric chloride in 10 ml. distilled water, and store in a stoppered brown bottle. Prepare fresh reagent weekly, and shake thoroughly before use.

4.4.1.3 *Alph-naphthol standard solution.*—Dissolve 0.10 gm. of alpha-naphthol in 100 ml. of the mixed solvent (see 4.4.1.1). Prepare a fresh solution every 2 weeks and store in a stoppered brown bottle.

4.4.1.4 *Procedure.*—Weigh 0.5 gm. of the specimen into a stoppered 50-ml. flask. Add 25 ml. of mixed solvent (4.4.1.1) and shake until the specimen is in solution or uniformly suspended in the solvent. Pipette 1 ml. of the reagent (4.4.1.2) into the flask and shake for 3 minutes. Pour the contents of the flask into a clean, dry, 6-inch by ⅝-inch test tube. Stopper the tube and allow to stand until the turbidity decreases to a minimum (approximately 5 minutes). Next, prepare a color standard by dissolving 0.5 gm. of alpha-naphthol-free thickener in 25 ml. of mixed solvent (4.4.1.1). Pipette 1.0 ml. of the alpha-naphthol standard solution (4.4.1.3) and 1 ml. of reagent (4.4.1.2) into the flask and shake for 3 minutes. Pour the contents of this flask into a test tube as above, stopper, and allow to stand until the turbidity decreases to a minimum. Compare the colors of the clear, supernatant liquid in the two test tubes.

4.4.2 *Moisture.* — The moisture content shall be determined by either of the following methods; however, in case of dispute the procedure specified in 4.4.2.1 shall govern.

4.4.2.1 *Procedure.*—Weigh out rapidly, to the nearest decigram, approximately 25 gm. of the specimen and immediately disperse in 500 ± 10 ml. of benzene in a roundbottom liter flask. After the sample has been dispersed (approximately 5 minutes), determine the moisture content in accordance with method 300.1.5 of Specification VV–L–791 with the following exceptions and additional details:

(a) Use the entire 500 ml. of suspension instead of the 100 ml. specified.

(b) Heat the flask by means of a liquid bath or an electric heating mantle. Maintain the temperature of the bath or heating mantle at 140 ± 5° C. during the distillation.

(c) Distill for 2 hours. Wash down the condenser tube with three 10-ml. portions of benzene, scrubbing the tube at the same time with a stiff brush to loosen water droplets. Resume the distillation for an additional ½ hour; then repeat the washing and scrubbing. Remove the condenser and loosen the water droplets adhering to the side wall of the trap by policing down with a glass rod equipped with a rubber policeman which has the end sealed. Form a good meniscus on the water collected in the bottom of the trap by wiping the surface of the trap wall at the interface level, exerting a firm pressure on the trap wall. Free any droplets of benzene caught in the water phase with the police-

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man. Remove water adhering to the policeman by twirling the policeman in the upper benzene phase.

(d) Protect the top of the condenser tube by a calcium-chloride tube or equivalent, except during the rinsing and scrubbing operations.

(e) At all times during the distillation, maintain the heating medium at a level about ¼ inch below the level of the suspension in the liter flask.

(f) Make a blank determination by subjecting 500 ml. of benzene to the same treatment as the suspension and correct all determinations accordingly.

(g) Determine the efficiency of recovery by treating similarly 500 ml. benzene to which 0.50 ml. water has been added. If, after correction for the blank, the apparatus fails to recover 0.50 ± 0.05 ml. of water, tighten all connections and repeat the test. No determination shall be made with apparatus which has not demonstrated satisfactory recovery.

(h) Ground glass connections may be used in the apparatus in place of the tight-fitting corks. Care shall be exercised to prevent the entrance of atmospheric moisture by condensation on such joints.

4.4.2.2 *Vacuum-drying method.*

4.4.2.2.1 *Apparatus.*—The apparatus shall be of the type shown in figure 1.

4.4.2.2.2. *Procedure.*—Tare the sample tubes with an identifiable, tightly-fitting, rubber stopper in place. Transfer sufficient sample to the tube to fill it to a depth of approximately 1½ inches. Close the tube with the stopper and reweigh. Mount the tube containing the specimen on a manifold outlet and immerse the tube to a depth of approximately 3 inches in a constant temperature bath maintained at 50 ± 0.5° C. Evacuate to a pressure of 3 millimeters of mercury and maintain for a period of 4 hours. Then gradually release the vacuum and remove the sample tube from the manifold. Close the sample tube with the rubber stopper used during the original weighing, wipe the tube dry, and then rinse with acetone. Allow to stand for approximately 5 minutes before weighing. Calculate the loss in weight as percent moisture.

4.4.3 *Granulation.*—Nest a No. 6 in a No. 40 U. S. Standard sieve conforming to Specification RR–S–366 and place a receiving pan at the bottom. Place 50.0 gm. of material in the top sieve, cover, and assemble in a mechanical shaker geared to produce 300 ± 15 gyrations, and 150 ± 10 taps of the striker per minute. Vibrate for 5 minutes with the tapper in operation. Weigh the portions retained on each sieve and calculate the amounts on a percentage basis.

4.4.4 *Sulfate.*—Weigh to the nearest mg., approximately 5 gm. of the specimen into a 400-ml. beaker. Add 25 ml. of concentrated hydrochloric acid and heat to effect solution. Dilute with water to approximately 200 ml. and add 10 ml. of bromine water. Boil until the solution clears and allow to cool. When cool, filter through No. 40 filter paper and collect the filtrate in a 600-ml. beaker. Wash the residue three times and add the washings to the filtrate. Heat the filtrate to boiling and add 25 ml. of a 15-percent barium chloride solution dropwise and 5 drops of 1-percent agar-agar solution. Allow to cool to room temperature and filter thrugh a tared Gooch crucible. Ignite for approximately 1 hour at a dull red heat (600° C.) in a muffle furnace. Cool to room temperature in a desiccator and weigh. Calculate the percent of sulfate as follows:

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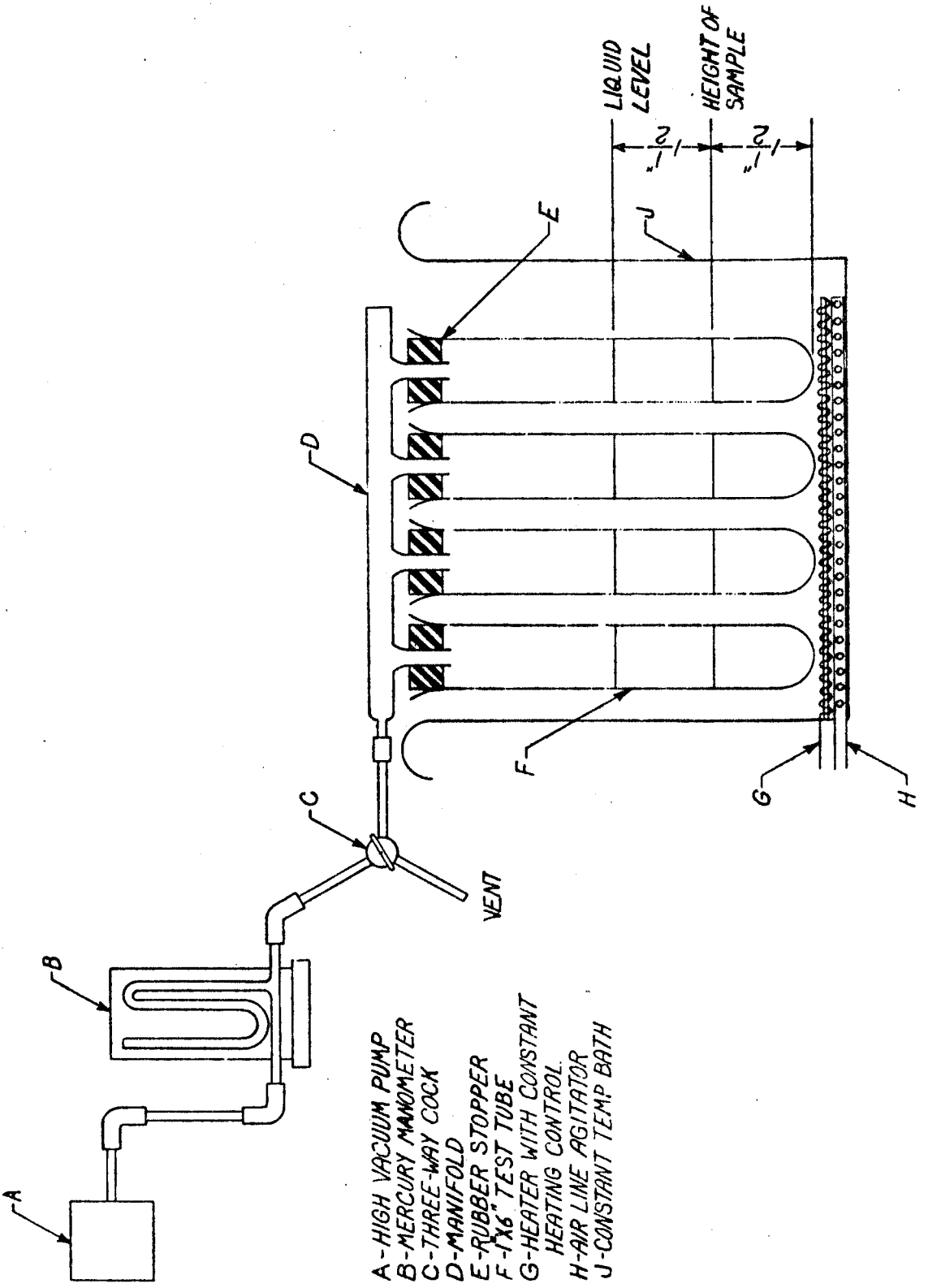


FIGURE 1.—*Vacuum drying apparatus.*

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$$\text{Percent sulfate} = \frac{41.15 \times A}{B}$$

where:

A = weight of $BaSO_4$ (gm.)
B = weight of sample (gm.)

4.4.5 *Gel formation.*

4.4.5.1 *Apparatus.* — The apparatus required for preparing gels for test purposes shall consist essentially of the following:

(a) A mechanical stirrer rotating at a speed of 300 ± 10 r.p.m. The stirrer shall be fabricated of a glass or corrosion-resistant metal rod, ⅜-inch in diameter, with four vanes of the same material as the rod, ⅞ by ⅞ by ⅛ inch, set at right angles and with the faces parallel to the axis of the rod.

(b) A wide-mouth, square-type quart jar approximately 3½ x 3½ x 7 inches.

(c) Storage containers of the type described in method 100 of Specification JAN–I–711.

(d) Constant temperature baths (air or liquid) capable of maintaining the specified test temperatures.

(e) A weighing room maintained at 70 ± 10° F. and a maximum relative humidity of 50 percent.

4.4.5.2 *Eight-percent gel.*—Stir 24.0 ± 0.2 gm. of the thickener into 276 ± 0.2 gm. of test solvent conforming to Standard MS602. Maintain the temperature of the solution at 77 ± 2° F. during the stirring period. Record the time required for disappearance of the vortex around the stirring shaft, and examine the gel for uniformity and freedom from lumps. The gel may be used for test in 4.4.6.3.

4.4.6 *Gel consistency.*

4.4.6.1 *Procedure.*—Prepare duplicate gels for each of the two specimens of the concentrations specified in 4.4.6.2 and 4.4.6.3, using the equipment specified in 4.4.5.1.

4.4.6.2 *Four-percent gel.*—Stir 12.0 ± 0.2 gm. of the specimen into 288 ± 0.2 gm. of test solvent conforming to Standard MS602. Maintain the temperature of the solution at 77 ± 2° F. during the stirring period. Place each gel in a separate storage container, and store for 24 ± 4 hours at 75 ± 10° F. After this storage period store both gels for an additional 2 to 4 hours at 77 ± 2° F. Then determine the consistency of each of the gels in accordance with method 100 of Specification JAN–I–711. Results of these two determinations shall agree within 25 gm.; if not, repeat this procedure until this condition is met, using fresh gel preparations.

4.4.6.3 *Eight-percent gel.*—Stir 24.0 ± 0.2 gm. of the specimen into 276 ± 0.2 gm. of test solvent conforming to Standard MS602. Maintain the temperature of the solution at 77 ± 2° F. during the stirring period. Place each gel in a separate storage container, and store for 24 ± 4 hours at 150 ± 5° F. After this storage period store both gels for an additional 2 to 4 hours at 77 ± 2° F. Then determine the consistency of each of the gels in accordance with method 100 of Specification JAN–I–711. Results of these two determinations shall agree within 50 gm.; if not, repeat this procedure until this condition is met, using fresh gel preparations.

4.5 Rejection and retest.—If any sample fails to meet any requirement of this specification for which it has been tested, the lot represented shall be rejected. Lots rejected for failure to comply with 3.2 may be reworked by dissolving in a caustic solution and blending into current production. Lots rejected for failure to comply with 3.3 through 3.7 may be reworked and resubmitted for acceptance or blended into current production. All lots containing reworked material shall be tested for conformance to this specification, when submitted for acceptance.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Thickener shall be pack-

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hundred-pound steel drums.—Unless otherwise specified in the contract or order, the 15¾- and 100-pound steel drums require no overpacking.

5.3 Marking.

5.3.1 *Five and one-quarter-pound can.*—Each can shall be stenciled with the following: Thickener, M1; Lot No.; Date of Manufacture; Manufacturer's Symbol or Name; and in letters approximately one-half inch high, "DO NOT OPEN EXCEPT FOR USE OR INSPECTION."

5.3.2 *Shipping containers.*—Shipping containers shall be marked in accordance with Standard MIL–STD–129. In addition, the tops of the containers shall be marked in 2-inch letters of a contrasting color: "KEEP DRY." All shipping containers, except the wood box used for the five and one-quarter pound cans, shall be marked in contrasting color with the following words of the specified sizes:

DO NOT OPEN EXCEPT FOR USE OR INSPECTION (1 inch).

MIXING INSTRUCTIONS INSIDE (1 inch).

Additional marking shall include:

Federal Stock Number:_____
Chemical Corps Stock Number:_____ (see 6.7)
THICKENER, M1

5.3.3 *Instruction sheet.*—The instruction sheet shown on figure 2 shall be made of yellow or white paper (8 by 10½ inches) conforming to Specification UU–P–641. The instructions shall be printed with black printer's ink and, after printing, the paper shall be folded as shown on figure 2.

6. NOTES

6.1 Intended use.—The material covered by this specification is intended for use in the preparation of thickened fuels for use in incendiary weapons and munitions.

6.2 Ordering data.—Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Size and type of container required (see 5.1).

6.3 Thickener meeting the requirements of this specification has been made satisfactorily using the following formulas shown in table II. The composition of the constituents of the organic acids mixture is shown in table III.

TABLE II.—*Formula for thickeners.*

Formula	Constituents	Parts by weight
1	Organic acids mixture:	
	Coconut fatty acids.	50
	Naphthenic acids.	25
	Oleic acid	25
	Aluminum content	5.4 to 5.8 percent
2	Organic acids mixture:	
	Coconut fatty acids.	30
	Naphthenic acids.	5
	Oleic acid	65
	Aluminum content	4.3 to 5.3 percent

6.4 Gasoline conforming to Specification VV–G–109 and having an aniline point not less than 46° C. nor more than 49° C., when tested in accordance with Specification VV–L–791, method 360.1.2 (benzol conforming to Specification VV–B–231 may be added in quantities necessary to bring the aniline point within range) may be used as a secondary standard in the consistency tests; however, each drum of gasoline must be standardized using test solvent conforming to Standard MS602 as the primary standard at

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aged in 5¼-pound cans, 15¾-pound steel drums, or 100-pound steel drums, as specified (see 6.2).

5.1.1 *Five and one-quarter-pound can.*—Five pounds, 4 ounces ± 1 ounce of thickener shall be packaged in a tear-strip can specified by Drawing D99–17–1; or in a rectangular can, 8½ by 5½ by 9 inches deep having a circular opening in the top approximately 4½ inches in diameter, fabricated of material conforming to Specification QQ–T–191. After filling, the circular opening shall be sealed by soldering. The exterior surface of the container shall be coated with lacquer or enamel conforming to Specification JAN–L–73, JAN–E–74, or MIL–E–10687. The color shall conform to color No. 3412 of Specification TT–C–595. The interior surface of the container shall be coated with a clear phenolic or vinyl coating.

5.1.2 *Fifteen and three-quarter-pound steel drum.*—Fifteen pounds, 12 ounces ± 4 ounces of thickener shall be packed in a 5-gallon drum, with welded side seams, conforming to Specification 37D of Regulations for Transportation of Explosives and Other Dangerous Articles, etc. The drum shall be constructed of 22-gage steel, or heavier, with full opening head. The head shall be provided with a 16-lug closure, with either a tubular rubber or flowed in gasket under the curl. All metal parts of the container shall be pretreated in accordance with Specification JAN–C–490, grade I or grade II, type 5. If grade I pretreatment is used, the interior of the drum and underside of the cover need not be further coated. If grade II, type 5 pretreatment is used, the interior of the drum and underside of the cover shall be coated with a protective coating which shall pass the water resistance test specified in Specification JAN–C–490. The outside of the drum and topside of the cover shall be coated with one coat of lacquer or enamel conforming to Specification JAN–L–73, JAN–E–74, or MIL–E–10687. The color shall conform to No. 3615 of Specification TT–C–595. The instructions for mixing (see figure 2) shall be inserted into a waterproof envelope (4¼ by 5¾ inches). The envelope shall be affixed with suitable adhesive to the inside of the cover prior to closure.

5.1.3 *One hundred-pound steel drum.*—One hundred ± 1 pounds of thickener shall be packed in a 30-gallon, fully-removable-drum of 2-gage steel, or heavier, with welded side seams, conforming to Specification 37D of Regulations for Transportation of Explosives and Other Dangerous Articles, etc. The cover shall have a flowed in or a continuous solid rubber gasket (Strip sponge rubber gaskets are not acceptable.) The closure ring lugs shall be of a heavier gage than the ring and shall be welded to the ring. If spot welded, a minimum of four welds will be required. Before filling and closing, the metal surfaces of the drum shall be prepared and coated and the instructions for mixing shall be attached as specified in 5.1.2. The drum shall be closed by applying a minimum load of 1,200 pounds uniformly around the periphery of the inner edge of the cover lip. The bolted ring closure shall then be securely tightened and a hexagonal lock nut shall be used to prevent the primary closing nut from vibrating loose. The lugs of the closure ring shall be separated by a gap of not less than ⅛-inch after tightening.

5.2 Packing.

5.2.1 *Five and one-quarter-pound can.*—Six cans of thickener packaged as specified in 5.1.1. shall be packed in a nailed wood box conforming to Specification JAN–P–106, with interlocking half-slotted partitions of double-faced corrugated fiberboard as interior packing. The inside dimensions of the box shall be approximately 17¼ by 17½ by 9 inches high. The box shall be strapped as specified in the appendix to Specification JAN–P–106. The instructions for mixing (see figure 2) shall be inserted into a waterproof envelope (4¼ by 5¾ inches). The envelope shall be affixed between the cleats of one end of the wood shipping box.

5.2.2 *Fifteen and three-quarter and one-*

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TABLE III.—*Composition of organic acids mixture constituents.*

Type of acid	Coconut fatty	Oleic	Naphthenic
Acid number (mg. of KOH)	250 to 280	188 to 209	200 to 245
Iodine number	20 (max.)	85 to 95	11 (max.)
Titer (°C.)	20 to 30 (should be sharp)	5 to 15	----
Moisture, insoluble, and unsaponifiable (percent).	2.5 (max.)	2.5 (max.)	20 (max.)
Ester number	3 (max.)	3 (max.)	----
Iron (percent)	0.01 (max.)	0.01 (max.)	0.02 (max.)
Rosin acids as abietic acid (percent)	----	1.00 (max.) (see 6.8)	----

intervals not exceeding one week. The correction factor will be applied in the consistency determination.

6.5 Batch.—A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

6.6 Gel stability.—The following is furnished for the information of the contractor: Impurities in the thickener which may cause oxidation of the thickener or partial or complete breakdown of gels formed with gasoline include excess water, lime, caustic soda, soaps of sodium, copper, lead, iron, manganese, and cobalt, powdered or sheet zinc or lead, lead nitrate, rust preventives containing sulfonates, amines, alcohols, and all acids. Tetraethyl lead, on the other hand, has no deleterious effects.

6.7 Stock number.—The Chemical Corps stock numbers for M1 thickener are as follows:

	Stock No.
(a) Five and one-quarter pound container	135215
(b) Fifteen and three-quarter pound drum	135219
(c) One hundred-pound drum	135216

6.8 Rosin acids content.—The following method is used to determine the rosin acids content of oleic acid as shown in table III: Place 40 gm. of the dry fatty acid into a 300-ml. round-bottom flask. Add a boiling chip, 100 ml. of anhydrous methanol, and slowly add 5 ml. of concentrated sulfuric acid. Reflux for 10 minutes, remove the flask, and cool to room temperature. Swirl and pour the flask contents into a separatory funnel which is three-quarters filled with a 10-percent sodium sulfate solution. Add 100 ml. of ethyl ether, shake vigorously, allow to settle, and draw off the salt layer. Repeat washing twice with more salt solution. (The last washing should not react pink to methyl orange.) Remove most of the water and salt solution and draw off the fatty layer into an Erlenmeyer flask. Add 20 ml. of absolute ethanol and 1 ml. of 1-percent phenolphthalein indicator to the flask. Titrate to the appearance of red color using approximately 0.3N sodium hydroxide. Calculate the percent abietic acid as follows:

$$\text{Percent abietic acid} = 1.058\,\frac{(A \times B \times 30.2)}{(C)} - 0.72$$

where: A = ml. of sodium hydroxide solution.

B = normality of sodium hydroxide solution.

C = weight of sample (gm.).

6.9 Sampling and testing.—When the contractor consistently produces high-quality

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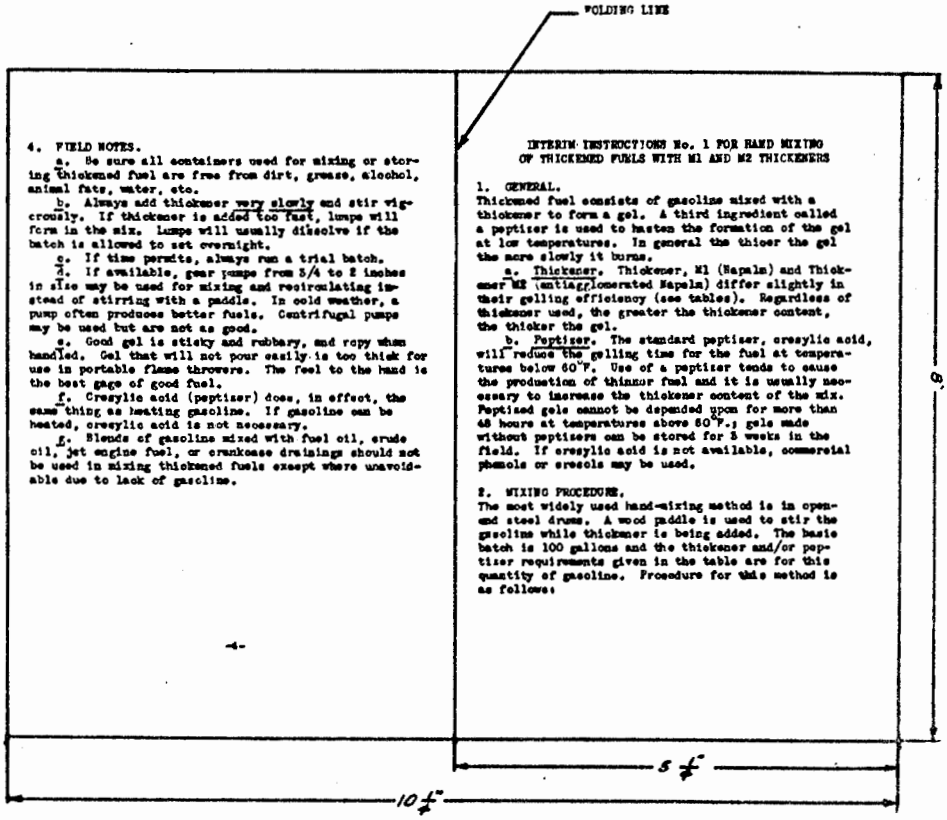
material and operates under a system of quality control acceptable to the Government, the Government, at its discretion, may modify, in whole or in part, the sampling and testing procedures specified herein. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures specified in this specification.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

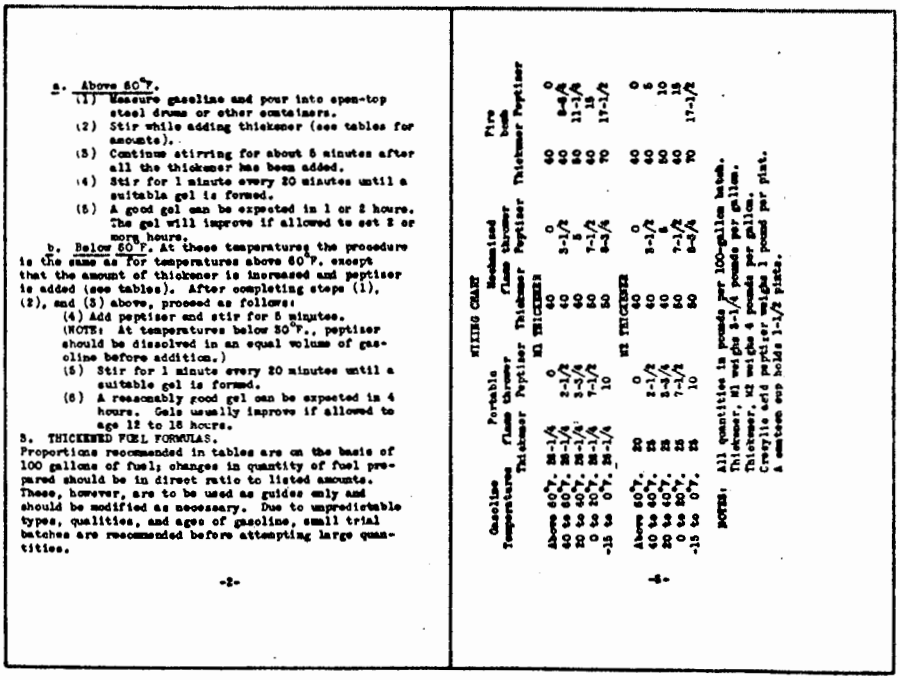
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Army—Chemical Corps
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OUTSIDE VIEW BEFORE FOLDING



INSIDE VIEW BEFORE FOLDING

FIGURE 2.—*Instructions for hand mixing thickened fuels.*

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